

UTAC v2.0: Executive Summary

Universal Threshold Activation Criticality

Domain-Specific β -Clustering and the Computational Criticality Universality Class

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The Problem: Predicting Catastrophic Transitions

Complex systems—climate, AI, ecosystems, financial markets, pandemics—undergo sudden, catastrophic transitions between stable states. Examples include:

- **Climate:** Atlantic Ocean circulation (AMOC) collapse, ice sheet disintegration
- **AI/Technology:** Large Language Model emergence (GPT, Claude), financial crashes
- **Health:** Neurodegenerative disease onset (Huntington's, ALS), pandemic tipping points
- **Ecology:** Ecosystem collapse, coral reef bleaching events

Critical Question: Can we predict *when* and *how fast* these transitions occur?

The Solution: UTAC Framework

The Universal Threshold Activation Criticality (UTAC) framework quantifies transition steepness via parameter β :

$$S(R) = \frac{1}{1 + e^{-\beta|R-\theta|}}$$

What β tells us:

- **Low β (≈ 4):** Gradual transition, years of warning (AI, markets)
- **High β (≈ 11):** Catastrophic transition, weeks/months warning (climate)
- **Extreme β (≈ 13):** Essentially no warning (protein diseases, ice sheets)

Five Key Findings

Empirical Validation

Domain	N	$\dot{\beta}$	$\Phi^{n/3}$ Pred.	Examples
Informational	27	4.5 ± 0.9	$\Phi^3=4.24$	LLMs, neuronal avalanches, markets
Geophysical	10	4.6 ± 0.8	$\Phi^3=4.24$	Earthquakes, volcanic eruptions
Biological	18	7.4 ± 0.9	$\Phi^4=6.85$	Microbiome, ecosystems
Climate	10	11.0 ± 1.0	$\Phi^5=11.09$	AMOC, ice sheets, permafrost
Neurodegen.	20	13.0 ± 1.8	$\Phi^{5.5}=13.8$	Huntington's, ALS, Alzheimer's
TOTAL	78	8.3 ± 4.1	—	Multi-modal distribution

Data Sources: Peer-reviewed literature (Nature, Science, PNAS, PRL), Impact Factor 5.0, full dataset at Zenodo DOI 10.5281/zenodo.17472834.

Three Actionable Policy Recommendations

1. Global β -Monitoring Network

Establish real-time UTAC tracking for high-risk systems:

- **Climate:** AMOC strength, ice mass balance (GRACE satellites), permafrost thaw
- **AI/Technology:** LLM capabilities, financial market coupling, epidemic spread
- **Health:** Protein biomarkers (neurodegenerative diseases), pandemic surveillance

Goal: Early detection of $R \rightarrow \Theta$ (approach to threshold) and β -spikes (acceleration).

2. Non-linear Steepness Adaptation (NSA) Policy Framework

Policy response must scale with β :

$$\text{Action Urgency} \propto \beta^2$$

- **Low β (< 7):** Standard monitoring, incremental policy
- **High β (7–10):** Enhanced surveillance, precautionary measures
- **Extreme β (> 10):** **Maximum preparedness**, assume transition imminent

Rationale: High- β systems offer weeks/months warning, not years. Waiting for "proof" means system loss.

3. Transdisciplinary Research Prioritization

Fund validation studies:

- **AI Safety:** Extract β from scaling laws (GPT-5, Claude 4, Gemini Ultra)
- **Consciousness:** Test Perturbational Complexity Index (PCI) $\approx \beta \approx 4.0$ hypothesis
- **Climate:** Validate high- β early warning systems, test cross-domain cascades (climate $\rightarrow$ geophysics)
- **Health:** Biomarker thresholds for neurodegenerative onset prediction

Why This Matters Now

Three Critical Systems Approaching Threshold ($R \rightarrow \Theta$):

1. **AMOC Collapse:** $\beta \approx 10.2$, projected 2025–2095, would trigger global climate reset
2. **West Antarctic Ice Sheet:** $\beta \approx 13.5$, 3–5 meter sea level rise, coastal civilization catastrophe
3. **AI Emergence:** $\beta \approx 4.2$, GPT-5/Claude-4 scale ($\sim 10^{13}$ params), unpredictable capability jumps

The field breathes in different rhythms. Understanding domain-specific β -dynamics is essential for navigating 21st century critical transitions.

Access Full Research

- **Main Paper (25 pages):** Domain-Specific Universality in Threshold Activation Criticality
- **Theoretical Extension (20 pages):** TAC Type-6 Implosive Genesis & Seismic Cascades
- **Full Dataset (78 systems):** CSV files with sources, β -values, uncertainties
- **Analysis Code:** R scripts for replication

Zenodo Repository: <https://doi.org/10.5281/zenodo.17472834>

Contact: Via Zenodo record or institutional collaborators

"If we wait for undeniable proof, the system may already be lost."

— UTAC v2.0 Framework